



华南理工大学

South China University of Technology

实验报告

课程名称: 《数字信号处理实验》

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学生学号:

学生专业: 人工智能

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Experiment 1

地点:	D3-B416	实验台号:	15
实验日期与时间:		实验教师:	宁更新
评分:			

1. Experiment Purpose

- 1) Understand the generation and analysis of common discrete-time signals.
- 2) Use MATLAB or Python to verify signal processing algorithms, plot results, and analyze observations.
- 3) Explain the main principles of the experiment and discuss how parameter changes affect the system response.

2. Experiment Content

2.1 Verification Experiment

1) Requirements and Tasks

Describe the task requirements, inputs and outputs, parameter settings, and expected observations.

2) Principle

Write the formulas used in the experiment here. For example, discrete convolution is defined as

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]. \quad (1)$$

3) Code

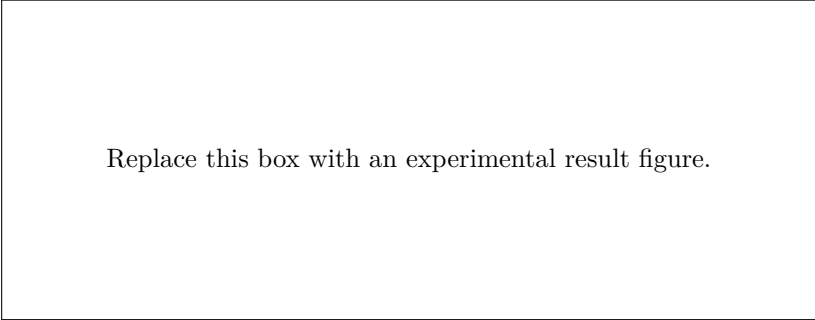
Listing 1 Discrete convolution example

```
h = [3 2 1 -2 1 0 -4 0 3];  
x = [1 -2 3 -4 3 2 1];  
y = conv(h, x);  
n = 0:length(y)-1;
```

```
stem(n, y);
xlabel('Time index n');
ylabel('Amplitude');
title('Output Obtained by Convolution');
grid on;
```

4) Result Analysis

Use figures, tables, or program output to explain whether the result matches the theoretical expectation. Discuss errors, boundary effects, and the influence of parameter changes.



Replace this box with an experimental result figure.

Figure 1 Example experimental result

2.2 Application Experiment

1) Requirements and Tasks

Describe the functional goal, design idea, input data, evaluation criteria, and expected output.

2) Principle

If the application task involves an algorithm, system block diagram, filter design, or spectral analysis, explain it here.

3) Code

Listing 2 Python code example

```
import numpy as np
import matplotlib.pyplot as plt

fs = 1000
t = np.arange(0, 1, 1 / fs)
x = np.sin(2 * np.pi * 50 * t)
```

```
plt.plot(t, x)
plt.xlabel("Time / s")
plt.ylabel("Amplitude")
plt.grid(True)
plt.show()
```

4) Results and Analysis

Present the program output, key figures, and analysis conclusions. It is recommended to discuss the results item by item according to the task requirements.

Table 1 Experimental result record

No.	Parameter	Result description
1	Parameter 1	Result 1
2	Parameter 2	Result 2

3. Discussion

Summarize what was learned from the experiment, including theory, implementation, debugging, and result analysis. Describe problems encountered, solutions, and possible improvements.

4. References

- [1] Oppenheim A V, Schafer R W. Discrete-Time Signal Processing. Pearson, 2010.
- [2] Proakis J G, Manolakis D G. Digital Signal Processing: Principles, Algorithms, and Applications. Pearson, 2007.